In Example 1.10, the first mode is unstable (both eigenvalues at $+1$) but controllable. The second mode is stable (eigenvalue at -2) but uncontrollable. Therefore, the system is stabilizable.

Definition 10 *If a system is observable or stable, then it is also detectable. If a system is unobservable or unstable, then it is detectable if its unobservable modes are stable.*

In Example 1.10, the first mode is unstable but observable. The second mode is both stable and observable. Therefore, the system is detectable.

Controllability and observability were introduced by Rudolph Kalman at a conference in 1959 whose proceedings were published in an obscure Mexican journal in 1960 [Kal60b]. The material was also presented at an IFAC conference in 1960 [Kal60c], and finally published in a more widely accessible format in 1963 [Kal63].

1.8 SUMMARY

In this chapter we have reviewed some of the basic concepts of linear systems theory that are fundamental to many approaches to optimal state estimation. We began with a review of matrix algebra and matrix calculus, which proves to be indispensable in much of the theory of state estimation techniques. For additional information on matrix theory, the reader can refer to several excellent texts [Hor85, Gol89, Ber05]. We continued in this chapter with a review of linear and nonlinear systems, in both continuous time and discrete time. We regard time as continuous for physical systems, but our simulations and estimation algorithms operate in discrete time because of the popularity of digital computing. We discussed the discretization of continuous-time systems, which is a way of obtaining a discrete-time mathematical representation of a continuous-time system. The concept of stability can be used to tell us if a system's states will always remain bounded. Controllability tells us if it is possible to find a control input to force system states to our desired values, and observability tells us if it is possible to determine the initial state of a system on the basis of output measurements. State-space theory in general, and linear systems theory in particular, is a wide-ranging discipline that is typically covered in a one-semester graduate course, but there is easily enough material to fill up a two-semester course. Many excellent textbooks have been written on the subject, including [Bay99, Che99, Kai00] and others. A solid understanding of linear systems will provide a firm foundation for further studies in areas such as control theory, estimation theory, and signal processing.

PROBLEMS

Written exercises

1.1 Find the rank of the matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

1.2 Find two 2×2 matrices A and B such that $A \neq B$, neither A nor B are diagonal, $A \neq cB$ for any scalar c, and $AB = BA$. Find the eigenvectors of A and

B. Note that they share an eigenvector. Interestingly, every pair of commuting matrices shares at least one eigenvector [Hor85, p. 51].

1.3 Prove the three identities of Equation (1.26).

1.4 Find the partial derivative of the trace of AB with respect to A.

1.5 Consider the matrix

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Recall that the eigenvalues of A are found by find the roots of the polynomial $P(\lambda) = |\lambda I - A|$. Show that $P(A) = 0$. (This is an illustration of the Cayley–Hamilton theorem [Bay99, Che99, Kai00].)

1.6 Suppose that A is invertible and

$$\begin{bmatrix} A & A \\ B & A \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

Find B and C in terms of A [Lie67].

1.7 Show that AB may not be symmetric even though both A and B are symmetric.

1.8 Consider the matrix

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

where a, b, and c are real, and a and c are nonnegative.

a) Compute the solutions of the characteristic polynomial of A to prove that the eigenvalues of A are real.

b) For what values of b is A positive semidefinite?

1.9 Derive the properties of the state transition matrix given in Equation (1.72).

1.10 Suppose that the matrix A has eigenvalues λ_i and eigenvectors v_i ($i = 1, \cdots, n$). What are the eigenvalues and eigenvectors of $-A$?

1.11 Show that $|e^{At}| = e^{|A|t}$ for any square matrix A.

1.12 Show that if $\dot{A} = BA$, then

$$\frac{d|A|}{dt} = \text{Tr}(B)|A|$$

1.13 The linear position p of an object under constant acceleration is

$$p = p_0 + \dot{p}t + \frac{1}{2}\ddot{p}t^2$$

where p_0 is the initial position of the object.

a) Define a state vector as $x = \begin{bmatrix} p & \dot{p} & \ddot{p} \end{bmatrix}^T$ and write the state space equation $\dot{x} = Ax$ for this system.

b) Use all three expressions in Equation (1.71) to find the state transition matrix e^{At} for the system.

c) Prove for the state transition matrix found above that $e^{A0} = I$.

1.14 Consider the following system matrix.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Show that the matrix

$$S(t) = \begin{bmatrix} e^t & 0 \\ 0 & 2e^{-t} \end{bmatrix}$$

satisfies the relation $\dot{S}(t) = AS(t)$, but $S(t)$ is not the state transition matrix of the system.

1.15 Give an example of a discrete-time system that is marginally stable but not asymptotically stable.

1.16 Show (H, F) is an observable matrix pair if and only if (H, F^{-1}) is observable (assuming that F is nonsingular).

Computer exercises

1.17 The dynamics of a DC motor can be described as

$$J\ddot{\theta} + F\dot{\theta} = T$$

where θ is the angular position of the motor, J is the moment of inertia, F is the coefficient of viscous friction, and T is the torque applied to the motor.

a) Generate a two-state linear system equation for this motor in the form

$$\dot{x} = Ax + Bu$$

b) Simulate the system for 5 s and plot the angular position and velocity. Use $J = 10$ kg m^2, $F = 100$ kg m^2/s, $x(0) = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$, and $T = 10$ N m. Use rectangular integration with a step size of 0.05 s. Do the output plots look correct? What happens when you change the step size Δt to 0.2 s? What happens when you change the step size to 0.5 s? What are the eigenvalues of the A matrix, and how can you relate their magnitudes to the step size that is required for a correct simulation?

1.18 The dynamic equations for a series RLC circuit can be written as

$$\begin{aligned} u &= IR + L\dot{I} + V_c \\ I &= C\dot{V}_c \end{aligned}$$

where u is the applied voltage, I is the current through the circuit, and V_c is the voltage across the capacitor.

a) Write a state-space equation in matrix form for this system with x_1 as the capacitor voltage and x_2 as the current.

b) Suppose that $R = 3$, $L = 1$, and $C = 0.5$. Find an analytical expression for the capacitor voltage for $t \geq 0$, assuming that the initial state is zero, and the input voltage is $u(t) = e^{-2t}$.

c) Simulate the system using rectangular, trapezoidal, and fourth-order Runge–Kutta integration to obtain a numerical solution for the capacitor voltage. Simulate from $t = 0$ to $t = 5$ using step sizes of 0.1 and 0.2. Tabulate the RMS value of the error between the numerical and analytical solutions for the capacitor voltage for each of your six simulations.

1.19 The vertical dimension of a hovering rocket can be modeled as

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{Ku - gx_2}{x_3} - \frac{GM}{(R + x_1)^2} \\ \dot{x}_3 &= -u\end{aligned}$$

where x_1 is the vertical position of the rocket, x_2 is the vertical velocity, x_3 is the mass of the rocket, u is the control input (the flow rate of rocket propulsion), $K = 1000$ is the thrust constant of proportionality, $g = 50$ is the drag constant, $G = 6.673E - 11$ m^3/kg/s^2 is the universal gravitational constant, $M = 5.98E24$ kg is the mass of the earth, and $R = 6.37E6$ m is the radius of the earth radius.

a) Find $u(t) = u_0(t)$ such that the system is in equilibrium at $x_1(t) = 0$ and $x_2(t) = 0$.

b) Find $x_3(t)$ when $x_1(t) = 0$ and $x_2(t) = 0$.

c) Linearize the system around the state trajectory found above.

d) Simulate the nonlinear system for five seconds and the linearized system for five seconds with $u(t) = u_0(t) + \Delta u \cos(t)$. Plot the altitude of the rocket for the nonlinear simulation and the linear simulation (on the same plot) when $\Delta u = 10$. Repeat for $\Delta u = 100$ and $\Delta u = 300$. Hand in your source code and your three plots. What do you conclude about the accuracy of your linearization?